16-bit Image-Process (for Windows, Ubuntu)
Exec example & files' iterator `Img16Prc<.exe>`.

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1. Developer's manual

1.1. The introduction

Console application `Img16Prc<.exe>` is demo-project to invoke `Img16Prc.so|.dll`. At the time of start, the SO|DLL shall locate in process C_{urrent} W_{ork} $D_{irectory}$ (aka ".\"). It also assumes possibility of few co-existing different binaries (named like `Img16PrcXYZ.dll` and having the same SW-interface) with same parameters and different functionalities for different application-domains .

SDK's files.

Binaries:

- `Img16Prc<.exe>` : iterator & demo console app;
- `Img16Prc.so|.dll` : the library.

Configuration:

- `Img16Prc*.ini` : initialization profile of `Img16Prc.so|dll`.

Sources:

- `Img16Prc.cpp` : source code of `Img16Prc<.exe>`;
- `..\I16_Cmn\I16_Intf.h,.cpp` : SW-interface of `Img16Prc.so|.dll`.

Solutions, projects:

- `*2008.sln, *2008.vcproj` : for MVS2008 (VC-Build toolchain);
- `*.sln, *.vcxproj` : for MVS>=2017 (MS-Build toolchain).

1.2. General flow of presentation-enhancement:

ModeId : processing mode; refers to section `[IPnn]` in INI-profile (normally that is file .\VAPrcFlu.ini).

```
----- Presentation ModeId ------------------    This sketch depicts dialog with
|                 ______________________________ |  single combo-box, which enlists
| The taxonomy : 00 "Abdomen supine"       v |  keys "sCapt=..." in sections
|                ----------------------------:-:|  [IPnn] (obtained e.g. via
|                : 01 "Abdomen prone"       : :|  "GetPrivateProfileSectionNames")
|                : 10 "Molars"              : :|
|                : 11 "Incisors"            : :|
|                : 20 "Ceramic PCB conveyer": :|
|                : 30 "Welding inspection"  : :|  End-User-App invokes
|                :__________________________:_:|  _Init( ..., ModeId ) <nearly>
-----------------------------------------------  once.
```

Adjustments : presents as integer composing of four 4-bit-factors "TBCS" (in hex format).
Factor "T" contains ad-hoc bit flags; factors "BCS" contain values 0..15.

```
Factors    : Bits : Alternate descriptions/interpretations/similarities
-----------:------:-------------------------------------------------------------
Tag        :12..31: ad-hoc instructions (e.g. "invert"; refer DLL-specific documentation).
Brightness : 8..11: gray_wnd-center, b/w-balance, gamma-correction.
Contrast   : 4.. 7: Global/Histogram Equalizer, gray_wnd-width.
Sharpness  : 0.. 3: Local/Spatial Equalizer, edge-enhancement, de-blur.
```

```
-- Presentation Adjustments --
|                0   8   15 |
| Brightness       ===I===  |
| Contrast         ===I===  |
| Sharpness        ===I===  |
-----------------------------
```

This sketch depicts tool-window with 3 sliders of range 0..15.

These values may change in each call _Invoke(..., Adj, ...).

1.3. Process-flow in demo-application:

Refer source code of _tmain(....) in "I16_Prc\I16Prc.cpp" and/or process-flow under debugger.

1) Parse command line.
2) Enlist source `*.raw` by wildcard; terminate if no such files.
3) Sort source `*.raw`.
4) Call `LoadLibrary`(...) (maps `VAPrcFlu.dll` into process' memory).
5) Call `_Init`(...) and allocate buffers.
6) For each source `*.raw`: read from disk; call `_Invoke`(...); save processed `*_?.raw`; (optionally) save `*_?.bmp`.
6) At once call `_Invoke`(...,nil,0,..) and save single summary frame (if one exist) as `*WxH.raw`.
7) Call `_Release`(...).
8) Call `FreeLibrary`(...).

Extended note: the list above omits (the necessary for maintenance of commercial End-User-App) dbg-txt-trace & dbg-files-dump.

The differences between demo-application & typical End-User-App

	Demo-application:	Typical End-User-App:
ModeId	Value from cmd-line.	Enlists sections [IP??] in VAPrcFlu.ini; choices '??' in ~dialog 'Presentation *ModeId*'.
Adjustments		Interactive UI changes of tags TBCS by ~tool -window 'Presentation *Adjustments*'.
_DbgDmp	Enables dbg-files-dump (for first *.raw<s>) or dbg-txt-trace (for all *.raws)	Logs dbg-txt-trace (via stderr; at least for the first frame in a data-series). Enables dbg-files-dump when special maintenance needs. Refer also topic **'Ad-hoc APIs I16_xxx : _DbgDmp'** below.

1.4. Execution of demo-application

This paragraph describes functionalities of initial `Img16Prc<.exe` >(which might be ~easily changed by other developers).

- Command-line parameters:

`WDT HGT wildcard Processing_mode TBCS <Flg>`: | Refer source code of _tmain(....)

```
  inp: "FileName.raw",                                  >in I16_Tst\Img16Prc.cpp and/or
  out: "FileName_NN_TBCS.raw".                          | process-flow under debugger.
- wildcard (inp frame<s>).
- Processing_mode (decimal integer).
- run-time adjustments "TBCS" (hex integer).
  Set Tag=0 (unless there is implementation-specific stipulation).
- Flg (hex integer; default "0").
```

- Optional parameter `Flg` (bitflags of `Img16Prc<.exe>`):

```
bit #0: (1) - save BMP<s>.
    #1: (2) - don't save RAW<s>.
    #2: (4) - don't save TBCS in files' names.
    #3: (8) - initially once invoke with "data_addr=nil" (out of FpS-meter).
 #4..7: (0) - no dbg;
      : (1) - dbg-txt-trace to stderr for all source *.raw<s>;
      : (2) - dbg-files-dump for inp *.raw #0,
      : (3) - for #0..1; (4) - for #0..2; etc...
#8..11: (0) - don't check AVX in CPUID,
    #0: (1) - demand arch:AVX,
    #1: (2) - demand arch:AVX2,
    #2: (4) - demand AVX512.
```

```
:\>Img16Prc 1500 1500 org.raw 01 111 20          [IP01], BCS='111',dbg-files-dump;
:\>Img16Prc 750 750 .\Dir\10??.raw 99 ABC 10     IP[99], BCS='ABC', only trace dbg-txt-trace to stderr;
:\>Img16Prc 1472 2172 .\Dir\*.raw 111 888        IP[111], BCS=888', normal workflow, no dbg.
```

1.5. Suggesting implementation stages (for End-User-App with interactive UI):

Stage "A".

Img16Prc.so|.dll at this stage might be only interface-stub (without relevant processing). INI-profile contains only section "[IP00]" with auto-generated keys. End-User-App calls _Init(..., 0), _Invoke(..., 0888).

Stage "B".

Img16Prc.so|.dll at this stage processes <less or more realistic> adjustments "TBC". Section "[IP00]" contains somewhat adjusted keys. End-User-App's UI implements controls from the "Presentation *Adjustments*" sketch and <when needs> check-boxes like "[x]Accuracy". End-User-App calls _Init(..., 0), _Invoke(..., 0888).

Stage "C":

Img16Prc.so|.dll at this stage processes <less or more realistic> adjustments "TBCS". INI-profile gets few sections "[IPnn]" (suggested by image-processing-experts). End-User-App 's UI implements controls from "Presentation *ModeId*" sketch. End-User-App calls _Init(..., nn), _Invoke(..., 0888), _Invoke(..., 0188), _Invoke(..., 0F88), etc.

Stage "D":

End-User-App's development ~completes. Image-processing-experts and developers of VAPrcFlu.dll proceeding Q&A. Interpretation of factors "TBCS" might change: e.g. "B" migh involve one of gray-window-center, gamma-correction or both of these two. Sections [IPnn] might add|remove keys (engineering UI to modify those keys might be implemented as ":\>rundll32.exe Img16Prc.dll,I16_Dialog" or as standalone executable).

2. Developer's reference

2.1. SW-interfacing with `Img16Prc.dll`:

- Refer available documents from specific DLL's project (such file<s> as `..\I16_Lib\readme.txt`).
- Include `I16_Intf.h` into a project. Pragma "`comment(lib,"Img16Prc.lib")`" statically links against `Img16Prc.dll`. For dynamic link include also `I16_Intf.cpp` into a project. Refer the comments at the top of that file.
- Invoke presentation enhancement following the sample code of "`_tmain`(...)" in `Img16Prc.cpp`. Consider the topics "**Suggesting implementation stages**" & "The differences between demo-application & typical End-User-App" above.

Extended note:

```
The identifiers' prefixes denote their scope:
  "t..." : a type,
  "c..." : a const,
  "r..." : a record's field (unforeseen change /out of sample illustrations/ highly-likely would cause side-effects; AKA ~"protected"),
  "m..." : a member-field (it was foreseen to be changed less or more freely; AKA ~"public"),
  "A..." : a parameter above stackframe-pointer (e.g. in Pleora eBus SDK this is "a..."),
  "q..." : quick|local  variable within stackframe (e.g. in Pleora eBus SDK this is "l..."),
  "x..." : a variable in the datasegment (e.g. in Windows SDK this is "g..."),
  "i..." : an intrinsic code of current module.
The identifiers don't mention their datatype (which shall be clear from a context).
```

Prefix "`x...`" denotes necessity of high attention (don't edit source code with such variable without whole project's correspondence|scope because of expecting side-effects). Prefix "`ix...`" denotes moderate level of concern (edit a code-snippet with correspondence|scope of current *.cpp).

Normally few different `Img16Prc*.so|.dll`s are developing in the ~same environment and share the same `I16F_Intf.h`. If End-User-App''s process occasionally needs different `Img16Prc*.so|.dll`s with different `I16_Intf.h` then one might:

step a) rename a set of files `I16_Intf.h,.cpp` (~ `I16a_Intf.h,.cpp`);
step b) modify routines' (& types' when necessary) identifiers' prefixes in renamed files (~ replace strings "I16_" to "I16a_").

2.2. Types and constants:

Basic types:

```
tI16_Inp : inp pixels ("short", "unsigned short").
tI16_Out : out pixels ("short", "unsigned short", "BYTE").
```

Ad-hoc constants:

```
Refer "const int cFl_...=..." for command-line's parameter Flg in _tmain( .... ) in "I16_Tst\Img16Prc.cpp".
```

2.3. Basic APIs I16_xxx():

Minimalistic functionality for ~"a dynamic viewer" (without a concern about threading).

```
int _Init( int W, int H, int ModeId )
- inits and allocates resources.
```

```
  WxH    : max image ROI's size in pixels (i.e. 'W*H' for "_Invoke( W, H, ... )").
  ModeId : the ModeId.
  return <0: error;
         =0: nothing to do;
         >0: adhoc flags (meaningful only for debug-trace; e.g. =1);
         >1 shl 18: mx possible output's dimension Wout or (Hout shl 16).

void _Release()
 - releases resources.
The code of Img16Prc assumes that summary stats (if ones exists) is already
returned|saved (by _Invoke(w,h,NULL,0,Dst,Adj)).  Specific caller might provide (via
_ParSet) additional afhoc flags here to manage further summary stats.


int  _Invoke( int w, int h, const tI16_Inp *Src, int wSrc, tI16_Out *Dst, int Adj )
- invokes process ModeId.
  w,h  : ROI_width, ROI_height.                    | the frame |
  Src  : input ROI's address;                      |Src*-----+ |   Dst*-----+
         in Img16Prc<.exe> the Src points          |   |  R  | |      |  R  |
         to the 'whole device                      |   h  O  | |      h  O  |
         frame'.                                   |   |  I  | |      |  I  |
  wSrc : input rows' stride;                       |   +--w--+ |      +--w--+
         Img16Prc<.exe> passes Wsrc=W.             +----wSrc---+
  Dst  : output buffer (of w*h).
  Adj  : the Adjustments.
  return <-63: fatal error (any other return code describes the output for specic
input);
         <0: non-fatal error;
         =0: nothing to do;
         >0: adhoc flags (meaningful only for debug-trace; e.g. =1);
         >1 shl 18: output's dimension as Wout or (Hout shl 16).
```

When `Src=nil, wSrc=0` then caller requests a *summary output* at given moment. `Img16Prc<.exe>` invokes this at once after wildcard's files (if no fatal error and there was a `return>0`).

2.4. Ad-hoc APIs I16_xxx():

The functions (unless there exist explicit implementation-specific stipulations) are:

```
void _DbgDmpReset()
       - resets dbg-files-dump (purges subdirectories ".\$*\*").

void _DbgDmp( const char *SubDir )
- SubDir="" minimizes debug-info; i.e.:
    disables dbg-files-dump; enables minimalistic dbg-txt-trace (to ~stderr).
- SubDir="MyName" switches debug-info ON; i.e:
    enables dbg-files-dump & verbose dbg-txt-trace.
    option 1: if SubDir[0]<>'$' then add prefix '$';
              GetCurrentDirectory(CWDKeep)+SetCurrentDirectory(CWDKeep\$MyName);
              dbg-files-dump ~".\XYZ.raw".
    option 2: GetCurrentDirectory(CWDKeep)+SetCurrentDirectory(CWDKeep\$WWWWxHHHH);
              dbg-files-dump ~".\MyNameXYZ.raw".
- SubDir=nil switches debug-info Off; i.e:
    SetCurrentDirectory(CWDKeep);
    disables dbg-files-dump & dbg-txt-trace.

Extended note:
The options 1,2 above depend on implementation of specific Img16Prc.dll.
For option 1:  _DbgDmp( "MyName" ); // CWD becomes ".\$MyName"
               _Invoke( ... );         // the library dumps files "XYZ.raw" to CWD
               _DbgDmp( NULL );        // CWD becomes original ".\"
```

```
For option 2:  _DbgDmp( "MyName" ); // CWD becomes ".\$WWWWxHHHH"
               _Invoke( ... );         // the library dumps files "MyNameXYZ.raw" to
CWD
               _DbgDmp( NULL );        // CWD becomes original ".\"

void _ParSet( int i, int v )
- adhoc_params[i] := v (no effect when 'i' is invalid);
- Img16Prc<.exe> doesn't call this procedure;
- for another application a I16-library could assume (as an example):
  entry #0 CUDA =-1:don't use, =0:auto, =1:mandatory
  entry #1 initialzaion profile =-1:don't use, =0:auto, =1:mandatory
                             or =0:use as default, 1:reset|overwrite from hard-
coding DSeg;
  entry #2 the min of retured err_codes (e.g. ~"don't save summary statistics").

int _ParGet( int i )
- returns adhoc_params[i] (returns 80000000h when 'i' is invald).

char* _ParStr( int i ) (optional)
- returns address of adhoc_string_params[i][256] (returns NULL when 'i' is invald).
If _ParStr(0)<>nil then Img16Prc<.exe> sets it  to:
- directory in the wildcard before _Init;
- input file' s name before _Invoke.
```